Are Investors Influenced by How Earnings Press Releases are Written?

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Abstract

Earnings press releases are an important means by which many firms communicate to investors. This study examines whether investors are influenced by how earnings press releases are written -- the tone and other stylistic attributes -- using actual earnings press releases and archival capital markets data in a standard short-window event study. To measure the tone and other stylistic aspects of press releases, I use elementary computer-based content analysis. Tone is measured using a frequency count of positive or negative words. Other stylistic aspects are the overall length of the press release, the overall percentage of numbers versus words, and the complexity of the words used.

Results suggest that tone of earnings press releases influences investors' reactions to earnings. An explanation for this result is provided by prospect theory (Tversky and Kahneman, 1981, 1986), which predicts that framing financial performance in positive terms, will cause investors to think about the results in terms of increases relative to reference points. Results also suggest that length and numerical intensity of the textual portions of earnings press releases may reduce the impact of unexpected earnings on market reaction.

Keywords: EARNINGS ANNOUNCEMENTS; TONE; CONTENT ANALYSIS; EVENT STUDY

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I. Introduction

Earnings press releases, though voluntary, are an important means by which many firms communicate to investors about their financial performance. FASB *Concepts Statement No. 1* (FASB 1978) specifically identifies news releases as examples of financial reporting. In 2003, the SEC added a new disclosure item to Form 8-K, formally requiring firms to file all earnings press releases.[1] Further, in July 2005 the SEC's securities offering reform, which significantly liberalizes communications around the time of public offerings, allows all issuers to continue publishing "regularly released factual business information…[for example] …an earnings release consistent with past practice, including the posting of and maintaining the release on an issuer's web site" (SEC 2005, 51). Although SEC Regulation G gives precise guidance on firms' disclosure of non-GAAP financial measures, other disclosure content in earnings announcements remains discretionary. In addition, the way such disclosure is written is discretionary, other than the requirement that it be written in "plain English."[2]

The purpose of this paper is to gain an increased understanding of the firm-investor communication process by examining whether investors are influenced by how earnings press releases are written: the tone and other stylistic attributes of press releases that accompany announcements of earnings. This study examines the impact of tone and other stylistic attributes of earnings press releases using 1,366 firm-year observations of annual press releases issued by firms in the telecommunications and computer industry between 1998 and 2002. To examine whether the behavior of capital markets shows an influence on investors of tone and other stylistic attributes, the study employs a classic short-window event study, a methodology widely used to examine factors determining the return-earnings relationship.

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[1] The new disclosure Item 12 "will bring earnings information within [the SEC's] current reporting system by requiring registrants to furnish to the Commission all releases or announcements disclosing material non-public financial information about completed annual or quarterly fiscal periods. New Item 12 does not require that companies issue earnings releases or similar announcements. However, such releases and announcements will trigger the requirements of Item 12." (SEC, 2003).

[2] The phrase "plain English" is used extensively in disclosure regulation. Revisions by the SEC (1998) to the Securities Act require plain English in key portions of registration statements and prospectuses, and the Sarbanes-Oxley Act of 2002 calls for real time disclosures in plain English.

Tone, defined in this study as the affect or feeling of a communication, is a function of both content and word choice. Thus, a more positive tone can be achieved by focusing on positive outcomes and/or by describing outcomes in a positive way. Any company with reasonably complex operations has a broad body of "news" from which to select which items it will report in the textual disclosure that accompanies its financial statements in a periodic earnings announcement. Consider, for example, a company choosing to report in the headline that the "Company Achieves Record Quarterly Revenues; Balance Sheet boasts nearly $1.1 billion in cash and marketable securities...," while the accompanying tabular income statement shows a loss per share of $3.54 compared to a year earlier earnings per share of $2.70 (*Business Wire* 2001).

Another way in which a firm can focus on positive outcomes is by selecting a benchmark that allows favorable period-to-period comparison, i.e. allows the firm to describe their performance as *better* compared to that benchmark. Research shows that strategically selected benchmarks do influence investors (Lougee and Marquardt 2004; Schrand and Walther 2000; Bowen, Davis and Matsumoto 2005). Experimental research demonstrates that two press releases presenting identical financial statements but accompanied by different textual explications, for example selectively emphasizing favorable comparisons, have a different impact on investors (Krische 2005; Brooke 2006). Other ways in which firms can focus on positive outcomes in earnings press releases include emphasizing a geographic, business or time segment that showed improvement. For example, consider a company's choice to highlight as a bulleted introductory point that the operating margin of one division "rises 0.7 percentage points to 29.7%," while the numbers in the accompanying tabular income statement reveal that the operating margin for the company as a whole fell by nearly one percentage point (Novartis Corporation SEC *Form 6-K*, January 20, 2006). Yet another way to present a positive press release to offer positive comments about future performance. Officer comments in press releases conveying positive (or negative) future expectations – apart from explicit earnings forecasts – have been shown to be significantly associated with market reaction (Hoskin et al. 1986; Francis et al. 2002).

Tone is also affected by word choice. Research on specific words in accounting narrative has examined annual reports rather than earnings announcements. Rutherford (2005), in an examination of annual report narratives as a corpus,[3] finds, for example, that upward directional words such as "increase" and "up" are far more prevalent than downward and that the language is generally biased toward the positive. Negative words in annual reports' letters from top management have been shown to be associated with market returns (Abrahamson and Amir 1996) and firm failure (Smith and Taffler 2000).

Although the disclosure content of earnings announcements is largely discretionary, firms differ in their ability to "manufacture" a positive tone depending on how their actual earnings compare to key benchmarks. These key benchmarks include: analysts' expectations, prior period results, and profit versus loss (Graham et al. 2005). This study therefore controls for firms' actual results relative to each of these benchmarks.

Other than tone, stylistic aspects that may influence investors include: quantity, textual complexity, and numerical intensity. Botosan's (1997) study of the relation between disclosure in annual reports and the cost of equity capital captures the quantity of disclosure (citing researchers' assumption that quantity and quality are positively related). However, Li (2006) shows that the length of an annual report, as well as its textual complexity, is negatively related to earnings and earnings persistence (though not to future market returns in following years.) Quantitative, i.e. numerical data in annual reports is considered to be more precise, useful and possibly credibility-enhancing and thus to affect the cost of equity capital (Botosan 1997). Similarly, greater precision in forecasts is viewed as enhancing investors' assessment of the credibility of management disclosure (Mercer 2004).

Computer-based content analysis is used to measure tone based on a frequency count of positive or negative words. This approach contrasts with previous accounting archival studies that specifically examine "tone" as a judgmentally assessed variable, without offering a formal definition of the word but instead relying on the generally understood meaning of the word as describing the optimism/pessimism or

[3] A corpus is defined as a "special collection of textual material collected according to a certain set of criteria" (Manning and Schutze, 1999, p. 119.)

positive/negative nature of the communication (Francis et al. 1994; Lang and Lundholm 2000). Other stylistic aspects include: quantity, measured as the overall length of the press release, i.e. word count; numerical intensity, measured as the overall percentage of numbers versus words; and textual complexity of the words used.

The key findings are as follows. Employing measures of stylistic attributes of earnings press releases alongside changes in accounting earnings as predictor variables for abnormal market returns provides evidence of higher event window returns as the tone of the press release becomes more positive. The impact of tone on market reaction is consistent with prospect theory (Tversky and Kahneman, 1981, 1986), which predicts that framing financial performance in positive terms, will cause investors to think about the results in terms of increases relative to reference points. Results also suggest that the effect of unexpected earnings on abnormal returns is diminished by the length and numerical intensity of press releases, and that textual complexity of the release suppresses the relationship between unexpected earnings and abnormal returns.

This paper offers the following contributions to the literature. First, the paper extends the stream of disclosure literature focusing on earnings press releases. Prior literature has examined the influence on investors of strategically-selected benchmarks and prospective comments in earnings press releases but has not examined the influence of tone and other stylistic attributes on investors. Prior literature has also examined the relation between certain words in, and overall readability of, annual report narratives and investor responses, but a similar examination has not been applied to earnings press releases and the market response to the release. Second, an explanation for the main results in this paper is provided by prospect theory (Tversky and Kahneman, 1981, 1986). Prospect theory has been widely applied in experimental research but has received limited attention in archival accounting research and has not, to my knowledge, previously been applied to the classic setting of archival capital markets research: the short window event study around earnings press releases. Third, this paper applies computer-aided content analysis to develop a numeric measure of the tone of earnings press releases. Prior research on tone of press coverage has relied on judgmental assessment, which limits replication of that research. By developing a numeric measure of tone, this study facilitates extension and replication.

The remainder of this paper is organized as follows. Section II develops hypotheses. Section III describes measurement of test variables including measurement of tone. Section IV provides descriptive statistics, and Section V presents results from the short-window event study. Section VI concludes.

II. Hypothesis Development

Are investors influenced by the way earnings press releases are written? The question is relevant to understanding market reaction to earnings announcements, an important means by which many firms directly communicate to investors about their financial performance.

An aspect of earnings press releases which makes the potential influence of their verbal components particularly relevant is the directness with which they are communicated to investors. A number of factors contribute to the direct communication links between firms and investors: firms' use of email or broadcast fax to distribute earnings announcements directly to investors, the online availability of press releases and SEC filings, and the implementation of Regulation FD.

The importance of earnings press releases in the firm-investor communication process is supported by anecdotal evidence, firm behavior, and regulatory actions. As anecdotal evidence, a former managing director of investor relations for Enron testified that analysts relied more on earnings releases than on SEC filings (*Wall Street Journal* 2006). Firms have demonstrated the importance of earnings press releases by increasing the quantity of disclosure in their releases. For example, the average word count of the earnings press releases examined by Francis et al. (2002) increased from 517 to over 2,400 between 1980 and 1999. Regulators' actions also evidence the importance of earnings press releases. As noted, FASB specifically identifies news releases as examples of financial reporting, and the SEC now formally requires firms to file all earnings press releases.

Despite the importance of earnings press releases, their content is largely unregulated. Regulation G gives precise guidance on firms' disclosure of non-GAAP financial measures, but other disclosure content in earnings announcements remains discretionary. In addition, the way such content is written is discretionary, other than the SEC's "plain English" requirement.

As a result of the broad discretion concerning earnings press releases, firms make many choices about the way their earnings press releases are written. This study examines whether the following attributes of earnings press releases influence investors: tone, quantity, textual complexity, and numerical intensity.

Tone

The tone of earnings news, as noted, is a function of both content and word choice. Earnings press releases are not devoid of tone. The 2005 securities offering reform, which significantly liberalizes communications around the time of public offerings, characterizes earnings announcements as "factual business information" (SEC, 2005); however, comparison of a statement considered to be hype, i.e. unduly positive, with a typical positive officer's comment in an earnings press release does not reveal an obvious distinction. For example, in the context of an SEC-enforced delay of a public offering due to comments by the CEO (prior to the 2005 securities offering reform), a former regulator explained that a statement such as "'we manufacture widgets'" is acceptable, but a statement such as "'our widgets are the best and demand for widgets is expected to double in five years'" is viewed as an attempt to excite investor interest, i.e. to hype a company's stock (*Wall Street Journal* 2004). The latter statement may be compared to a typical comment in an earnings press release. A typical positive officer comment appearing in earnings announcements (from Hoskin et al. 1986, 30) --"Looking further ahead Allen said that the company 'possibly' will double its revenue thus topping $2 billion in 1983" – is arguably similar to the statement considered to be hype.

As noted above, tone of an earnings press release can be affected in various ways, including: emphasis on positive aspects of performance, strategic selection of prior year benchmarks to allow favorable comparisons, and selection of specific language with greater positive affect. [4]

Strategic selection of prior year benchmarks in earnings announcements has been shown to influence investors, where investor reaction is measured using abnormal returns around the announcement date (Schrand and Walther 2000; Lougee and Marquardt 2004; Bowen et al. 2005). Schrand and Walther (2000) show that firms strategically select prior period's non-recurring earnings components as benchmarks, allowing favorable performance comparisons. Similarly, Lougee and Marquardt (2004) show that firms strategically elect to present pro forma earnings to allow favorable comparisons; further, firms are significantly more likely to present pro forma earnings when their GAAP earnings failed to increase relative to the prior period, or failed to exceed analysts expectations. Bowen et al. (2005) show that firms strategically emphasize metrics that allow them to *say* – in the headlines or text of earnings press releases – more *positive* things about their performance (Bowen et al. 2005). In summary, firms strategically choose and emphasize benchmarks that allow them to report an *increase* in some metric.

In strategically selecting and/or emphasizing prior year benchmarks, it is true that the firms report factual business information, but they also create a more positive tone of communication. Thus, strategic selection and emphasis on a particular benchmark can be considered one example of a more general phenomenon of creating a positive tone in reporting performance.

Other ways in which firms can focus on positive outcomes in earnings press releases include emphasizing a geographic, business or time segment that showed improvement, or offering positive comments about future performance. Comments by officers in press releases conveying positive (or negative) future expectations – apart from explicit earnings forecasts – have been shown to be significantly associated with market reaction (Hoskin et al. 1986; Francis et al. 2002).

[4]Such activities could be considered opportunities to pursue a "sanitization strategy" in which firms announce good news and suppress bad, modeled by Shin (1994).

Two archival accounting studies specifically examine "tone" as a variable, without offering a formal definition of the word but rather relying on the generally understood meaning of the word as describing the optimism/pessimism or positive/negative nature of the communication. Each of these studies judgmentally assesses the tone of disclosures, including both news released by the firm as well as news from other media sources. Francis et al. (1994), in an examination of market reaction to adverse earnings announcements, find no evidence of announcement day market returns being associated with the tone of press coverage in the year prior to the adverse announcement. In contrast, Lang and Lundholm (2000) note positive correlation (untabulated results) between market returns and the frequency of optimistic statements by companies in the 18 months prior to announcing seasoned common stock offerings; however, their coding rule that "absent some disclosed benchmark, earnings announcements are assigned a tone based on a random walk of quarterly earnings" suggests their results may be capturing market responses to quarterly earnings.

Tone is also affected by word choice. Accounting research on the influence on investors of specific words has examined chairman's letters in annual reports rather than earnings announcements, but has shown evidence of a relation between negative words in the letter and market returns (Abrahamson and Amir 1996) and firm failure (Smith and Taffler 2000). Abrahamson and Amir (1996) show a significant negative association between a firm's market-adjusted returns over the year following an annual report and the relative number of negative words in the chairman's letter. Although their focus is only on negative comments and is not on tone of the word choice per se, because they include factually undesirable words such as "loss" among their negative words, the results of their study are closely related to an examination of tone. Smith and Taffler (2000) show that key words and phrases from the chairman's letter are associated with subsequent firm bankruptcy.

In summary, accounting research on the influence of tone on investors is limited and inconclusive. Differences between this study and previous research may be noted. First, this study examines earnings press releases rather than annual report disclosures. Second, unlike studies of tone of press coverage, this study examines only earnings press releases authored by the firm, because the primary focus is on the communication process between firm and investor. Third, this study develops a numerical measure of the positive and negative tone of earnings press releases which will facilitate replication and extension of the analysis.

Research from related disciplines provides evidence of the influence of verbal components in situations analogous to financial reporting. Experimental research (Tversky and Kahneman 1981, 1986; Kahneman 2002) has shown that individuals' probability-dependent judgments are affected by the terms in which expected outcomes are expressed, i.e. "framing effects"; and prospect theory predicts that choices differ when outcomes are framed in positive versus negative terms in relation to some neutral reference outcome. For example, medical choices differ when outcomes are couched in terms of survival rates instead of mortality rates, (Tversky and Kahneman 1986). As another example, business investment decisions differ when outcomes are couched in terms of jobs and plants lost instead of jobs and plants gained (Bazerman 2002). This economic framing research focuses on radically different ways of describing outcomes rather than the style of communication, but is somewhat related to the issue of earnings press releases. For example, a particular change in net income could be described as increase in revenues or a decrease in expenses or both.

Overall, substantial evidence that readers are influenced by how information is written appears in analogous research in judgment and decision-making. This body of research, as well as the empirical research in accounting, motivates the hypothesis that the tone of the textual portion of earnings press release, as well as actual financial results, affects market reaction.

Hypothesis 1 (H1): The tone of earnings press releases, as well as actual financial results, affects investors' reaction to earnings announcements.

Quantity

Quantity of disclosure is often used as measure of disclosure quality on the grounds that regulation and litigation threats, as well as reputation effects, will assure that all disclosure is of high quality (Botosan 1997). In other words, more disclosure is assumed equivalent to better disclosure.

In contrast, some experimental accounting research has shown that disclosure of increased amounts of accounting information, i.e., notes in addition to financial statements, does not improve decision-making, possibly because of information overload (Casey 1980). This alternative view that excess disclosure diminishes investors' decision-making ability is evident in the Supreme Court's statement:

> "if the standard of materiality is unnecessarily low, … management's fear of exposing itself to substantial liability may cause it simply to bury the shareholders in an avalanche of trivial information - a result that is hardly conducive to informed decision making" (Choi and Pritchard 2003, n.225).

Li (2006) shows a negative relation between length of annual report narrative and earnings, but no relation with future stock returns over the following four years.

These conflicting views of disclosure quantity give rise to the hypothesis (non-directional) that the quantity of disclosure in the earnings press release affects investors' reaction to unexpected earnings.

> *Hypothesis 2 (H2): The quantity of disclosure in the earnings press release affects the relation between market reaction and unexpected earnings.*

Textual complexity

Textual complexity, or basic readability is a widely used measure of style. In financial disclosure, regulatory attention to textual complexity is apparent in the emphasis on "plain English," for example in the Securities Exchange Act of 1934, the Sarbanes-Oxley Act of 2002, and SEC rules (SEC 1998). A former SEC Chairman writes, "The SEC could do a lot more to improve disclosures by requiring companies to use plain English in their financial statements…it would be one of the most pro-investor steps the SEC could take to avoid future Enrons" (Levitt 2002, 157).

Research on the association between the textual complexity of a firm's annual report narrative and its earnings suggests that firms may opportunistically increase complexity to obscure adverse information from investors, but the complexity does not affect future market prices over the following years (Li 2006). To the extent that textual complexity in earnings announcement is perceived by investors as obfuscating adverse information, there could be some effect on investor reactions to unexpected earnings.

Hypothesis 3 (H3): The textual complexity of disclosure in the earnings press release affects the relation between market reaction and unexpected earnings.

Numerical intensity

Numerical intensity refers to the amount of quantitative information in financial disclosure. Botosan (1997) considers quantitative information to be of higher quality because of its precision and suggests that such information enhances management's credibility. Greater precision, as applied to forecasts, enhances investors' assessment of the credibility of management disclosure (Mercer 2004).

Decision-making research, although possibly not uniformly applicable to accounting settings, offers reasons for the preference of words over numbers in communicating generally: "Most people understand words better than numbers" and "numbers are perceived as conveying a level of precision and authority that people do not associate with their opinions" (Budescu et al. 1988, 281). Related to this observation, Wallsten et al. (1993, 177) find that, in communicating about uncertainty, people prefer to express their beliefs in words rather than numbers, but prefer to receive information numerically; however, the quality of decision "on average decision quality is roughly equivalent under the two modes."

Collectively, this research gives rise to the hypothesis (non-directional) that the relation between abnormal returns and unexpected earnings is affected by the numerical intensity of the earnings press release.

Hypothesis 4 (H4): The numerical intensity of disclosure in the earnings press release affects the relation between market reaction and unexpected earnings.

Relevance of investor sophistication

Miller and Bahnson (2002), who describe "hyping" and "spinning" as attempts by companies to mislead investors by focusing only on positive news, assert that such attempts cannot be successful because investors have the ability to gather private information from other sources and to evaluate public information correctly. Though the nature of the influence may depend on the relative sophistication of the investor, there are reasons to believe that both unsophisticated and sophisticated investors may be influenced by the verbal components of financial disclosure.

Unsophisticated investors may lack the ability to fully understand the information in press releases and thus fail to receive clear communication. Investors who lack the ability to fully understand the information in the press release may rely only on the portion of the press release with which they are familiar. Such an overweighting of the value of items with which an individual is familiar is an instantiation of the recognition heuristic examined by Gigerenzer (2001) in studies of judgment and decision-making. In experimental accounting research, Hofstedt (1972) presents evidence of less sophisticated investors placing greater reliance on the president's letter in annual reports when inconsistency existed between the tone of the report and the direction of change in EPS.

For sophisticated investors, the verbal components of press releases likely *complement* the financial statement information in all situations since such investors are less likely to be misled by lack of clarity or consistency. Warren Buffet, for example, has been quoted as saying he would not invest in a company whose disclosure (in this case, footnotes) he cannot understand, because he believes this would imply intentional obfuscation (Miller and Bahnson 2002). Disclosure tone and complexity may be interpreted as offering signals about intangible features such as management's credibility. Such signals might be recognizable to a sophisticated investor and useful in evaluating firm performance. Sophisticated investors may also have a better ability to understand longer disclosure and more numerically intense data.

The research discussed above motivates the following hypothesis.

> *Hypothesis 5 (H5): The market impact of the tone and other stylistic aspects of earnings press releases varies with investors' sophistication.*

III. Measurement of Test Variables

The measure of quantity of disclosure is length of the press release, i.e. a count of the total words in the earnings press release *(WORDS_AL)*. Textual complexity (*H3)* is measured as the number of characters per word *(COMPLEX),* and numerical intensity (*H4)* as the percent of the textual portions, i.e.

not including the financial statements, of press release that is numerical (numerical terms divided by total word count, *NUMERICAL*).[5] The proxy for sophisticated investors *(H5)* is the percentage of shares owned by institutional investors *(INSTITUTIONAL)* where a higher percentage would indicate greater sophistication.

The measure of tone *(H1)* is based on a frequency count of the number of positive and negative words, obtained using basic content analysis software (Diction 5.0 is available from dictionsoftware.com.) The words counted as positive and negative within earnings press releases are shown in Figure 1. *TONE* is calculated as the count of positive words minus the count of negative words divided by the sum of positive and negative word counts.[6] The maximum and minimum values of *TONE* are one and minus one, respectively. Separately, positive tone *(POSITIVE_TONE)* is defined as the frequency count of words on the list of positive words scaled by total words in the press release, and negative tone (*NEGATIVE_TONE*) is defined as the frequency count of words on the list of negative words, scaled by total words. The tone measure reflecting relative overall neutrality (*SCALED_TONE*) is measured as the difference between *POSITIVE_TONE* and *NEGATIVE_TONE.*

This approach to measuring tone may at first seem too simplistic to be useful, but similar metrics have been found to be useful in prior research. In accounting research, Abrahamson and Amir (1996) use a similar approach in analyzing the content of president's letters in annual reports – namely frequency counts of negative words divided by the total words in the letter. The authors examine only negative words because (a) their research approach, though computer-aided, requires substantial manual coding of text and is therefore extremely labor-intensive, and (b) they argue that positive words are so common that they would be unlikely to provide useful discriminatory power. Clatworthy and Jones (2003), in their study of impression management in chairman's statements, also use a combination of computer and manual analysis to classify news as good, bad or neutral depending on negative and positive word counts. Hussainey et al. (2003) use text analysis software and keywords such as "expect" and "will" to quantify

[5] Numerical terms include integers, numbers in lexical format such as "one", and terms referring to numerical operations (Hart 2000).
[6] This formula is presented in Uang and Taffler's (2005) study of auditor's opinion letters.

forward looking statements in their study of stock prices leading earnings. Smith and Taffler (2000) use frequency counts of key-words in CEO letters divided by total number of words in the letters to develop word and theme-based measurements of CEO letters (though not tone), and find that these measurements are predictive of bankruptcy. Uang et al. (2005) find that the tone of auditor's going-concern narrative, measured by the relative proportion of negative to positive keywords, predicts the severity of future outcomes such as bankruptcy.

Term frequency count metrics have also been found to be very useful in other domains. Using term frequency counts to capture the content of a document is commonly used for textual analysis in the social sciences, where such techniques fall into the general category of analysis referred to as content analysis (Neuendorf 2002). Additionally, term frequency counts are commonly used both for text-based information retrieval and for text categorization (Hand et al. 2001; Manning and Schutze 1999), two applications that fall into the overall category of automated language analysis, referred to in the computer sciences as natural language processing (NLP).[7]

When employing term frequency count metrics, two generic issues that must be addressed are synonymy and polysemy (lexical ambiguity).

The issue of synonymy results from the fact that similarity of terms is not captured in the term frequency count approach to text representation. To address the issue of synonymy, this study uses a thesaurus-based approach, which is appropriate given the focus only on corporate earnings press releases which cover a limited domain of discourse.[8] The thesaurus-based approach means that the lists of positive and negative words are created by including words with similar meanings.

[7] Hand et al. (2001, 457) note that representing texts as a bag of words is one basis for conceptual and practical distinctions between retrieval of text-based information (IR) and NLP. They define NLP as programming a computer to " 'understand' text data in the sense that it can map the ascii letters of the text into some well-defined semantic representation,." but "most practical IR systems in use today… rely on simple term matching and counting techniques, where the content of a document is implicitly and approximately captured (at least in theory) by a vector of term occurrence counts."

[8] Hand et al. (2001) indicate a thesaurus-based method is one way to overcome the issue of synonymy, a limitation of the term-vector approach to text representation.

The issue of polysemy in this study results from the fact that the affect of a word may differ depending on its context. Many of the words in the tone word lists are relatively unambiguously positive or negative. For example, it is reasonably clear that "delighted" and "excellent" are positive, while "disappointed" and "worst" are negative. However, the directional words, e.g., "increased" and "decreased", carry a particular potential for ambiguity. As an example, the word "increased" is included in the positive tone word list; but if an earnings press release states that some generally undesirable item increased, e.g., expenses increased, then the word "increased" could not be considered unambiguously positive in that context.

To determine whether an upward (downward) directional word is generally used in a positive (negative) sense, summary data is developed on the context of each directional word as it appears in the corpus of 1,366 press releases used for this study. The summary contextual data is examined to determine the percentage of occurrences of inherently desirable (undesirable) financial items, where desirable (undesirable) items are defined as those which are positively (negatively) related to increases in firm valuation in commonly used equity valuation models.[9] This approach is analogous to statistical disambiguation methods (Manning and Shutze 1999). Results of the analysis confirm that the upward (downward) directional words are, in the majority, consistent with a positive (negative) affect. Additional detail is provided in the appendix.

IV. Data and Descriptive Statistics

The sample in this study includes firms from the telecommunications and computer services industries, and related equipment manufacturers for the period 1998 to 2002.[10] During this period, there were close, notable linkages between telecommunications and computer services companies, implying a similar disclosure environment. The main rationale for examining this industry and time period is the

[9] Common equity valuation models are explicated in Stowe et al. (2002).
[10] (Fama and French 1997) define Telecommunications as sic codes 4800-4899; these SIC codes actually cover telecommunications, radio, television and cable. Related equipment manufacturers include SIC codes 3661-3669. The computer services industries include SIC codes 7370-7374 (programming, software, systems design, processing). Related equipment manufacturers include SIC codes 3570-3579. The source for the SIC code list is www.SEC.gov.

greater uncertainty created by the stock market boom of the late twentieth century. Research suggests that investors rely more heavily on non-financial information in periods of uncertainty and rapid technology change (Amir and Lev 1996).

The sample includes only companies with a fiscal year end of December and share price greater than $1, and is restricted to those companies for which data matched from the following three sources: (i) COMPUSTAT (source for report date and accounting data); (ii) CRSP (source for market data); and (iii) IBES (source for analysts' forecasts). In addition, because of the study's focus on firm-investor communication, the sample includes only companies whose earnings press release is available on Lexis-Nexis or Factiva. The company's direct press release is identified by source (typically on *PRNewswire* or *Business Wire*), by length (longer than non-company authored reports), and ultimately by inclusion of a company contact (usually an investor relations contact). The sample includes 1,366 firm-year observations for 562 firms.[11]

The basic event study reported here uses abnormal market returns around the earnings announcement dates as the dependent variable, with abnormal returns *(CAR)* defined as the accumulated return in excess of the CRSP equal-weighted market portfolio over the three event window days from -1 to +1 days around the earnings announcement date. Unexpected earnings *(UE)* is calculated as actual EPS minus EPS reported in the previous year, scaled by share price at the beginning of the period.

Table 1 provides descriptive statistics on each variable, and Table 2 provides a correlation matrix. Abnormal returns *CAR* are negatively (positively) associated with *NEGATIVE_TONE (POSITIVE_TONE*). Since *TONE* is higher when the positive tone metric exceeds the negative, it is also true that CAR is positively associated with these combined tone measures. Unexpected earnings *UE* are positively associated with positive tone and the combined tone metrics. Two findings -- the negative association between profitability *NIgt0* and length of earnings press release *WORDS_AL*, and between profitability *NIgt0* and textual complexity *COMPLEX* -- are comparable to findings by Li (2006) of a

[11] The sample is an unbalanced panel because of missing data in some years.

negative relation between a firm's earnings and the readability (length and textual complexity) of its annual report.

V. Results of short-window event study

This section presents summary results from the short-window event study. The first question is whether tone of the earnings press release, in addition to financial results, affects market reaction (*H1*). I estimate the following regression using 1,366 firm-year observations:

$$CAR = \alpha + \beta_1 UE + \beta_2 \ln MV + \beta_3 MBE + \beta_4 NI\_gt0 + \beta_5 TONE + \varepsilon \quad (1)$$

Where

CAR = cumulative abnormal returns from day t-1 to t+1, with day 0 as earnings announcement day,

UE = unexpected earnings.

lnMV = log of the market value of the firm's common equity,

MBE = an indicator variable equal to one if earnings exceed analysts' forecast,

NI_gt0 = an indicator variable equal to one if earnings are greater than zero,

TONE = tone measurement, the number of positive words minus the number of negative words, scaled by the sum of positive and negative words.

Since the data is a panel data set, all the regressions account for fixed company and year effects, i.e. include both group and time dummy variables. These effects capture constant firm-specific and time specific factors not explicitly capture in the regression equation. Such constant firm-specific effects may include, for example, firm's ritualistic disclosure behavior, defined by Gibbins et al. (1990) as passive, repetitive procedures.

Inclusion of the variable *lnMV* controls for firm size. Previous research shows that abnormal returns during earnings announcement periods are related to firm size (Ball and Kothari 1991, 719). Research also shows evidence of a difference in market response to firms reporting losses versus profits (Hayn, 1995), and the regression therefore includes a control variable indicating profitability *NI_gt0*.

A firm's ability to create a positive tone is presumably constrained by how its actual financial performance compares to commonly used benchmarks. Important benchmarks include prior year's results, profit versus loss, and analysts' forecast (Graham et al 2005). The variables *UE* and *NI_gt0* capture performance relative to the first two of these benchmarks, and the variable *MBE,* an indicator variable equal to 1 if the company beat the average of analysts' forecasts, captures performance relative to analysts' forecasts.

Table 3 presents results. As expected, both unexpected earnings and earnings in excess of analysts' expectations are associated with positive market reaction, as shown by the significant positive coefficients on *UE* (p=0.05) and *MBE* (p=0.00). The significant positive coefficient on *TONE* (p=0.02) provides support for hypothesis 1 that tone, in addition to actual financial results, affects market reaction to earnings announcements.

A question may arise whether *TONE* captures only an increased overall amount of disclosure. To address this question, the columns in Table 3 labeled [2] present coefficients from estimating the regression, but including an additional control variable for the length of the earnings announcement *WORDS_AL*. Results are robust to the inclusion of this control. The coefficient on *TONE* remains significant (p=0.02).

The impact of tone on market reaction is consistent with prospect theory (Tversky and Kahneman, 1981, 1986), which predicts that framing financial performance in positive terms, will cause investors to think about the results in terms of increases relative to reference points. To examine a further implication of prospect theory, namely whether market reaction to framing is convex above a neutral reference point (Tversky and Kahneman 1981), *TONE* is replaced with a measure of relatively neutrality of tone, *SCALED_TONE,* along with the square of that term. Results are shown in the columns labeled [3] in Table 3. The positive coefficient on *SCALED_TONE* and the negative coefficient on the square of that term, $SCALED\_TONE^2$, show that the effect of tone on market reaction is positive and convex. This can be interpreted as an indication that market reaction increases as tone becomes more positive, but only up to a point.

Tables 4 presents results of a regression addressing the market impact of stylistic attributes other than tone. The regression is estimated using the following equation:

$$CAR = \alpha + \beta_1 UE + \beta_2 \ln MV + \beta_3 MBE + \beta_4 NI\_gt0 + \beta_5 TONE + \beta_6 WORDS\_AL + B_7 NUMERICAL + B_8 COMPLEX + \varepsilon \quad (2)$$

Where variables are as defined above, and additionally

WORDS_AL = total word count of the press release

NUMERICAL = numerical intensity, defined as terms in the text portion of the press release scaled by total words

COMPLEX = textual complexity, defined as characters per word.

Results do not indicate that any of the stylistic attributes, have a significant effect on market reaction. To test whether these stylistic attributes individually affect the relation between market reaction *AR* and unexpected earnings *UE*, as suggested by hypotheses 2,3, and 4, interaction variables are used. Interaction variables calculated as the product of *UE* and the stylist variable are included as follows:

$$CAR = \alpha + \beta_1 UE + \beta_2 \ln MV + \beta_3 MBE + \beta_4 NI\_gt0 + \beta_5 TONE + \beta_6 STYLE* + B_7 STYLE*INTERACTION\_UE + \varepsilon \quad (3)$$

Where variables are as defined above, and additionally

*STYLE** is alternately length *WORDS_AL*, numerical intensity *NUMERICAL*, or textual complexity *COMPLEX*, and

*STYLE*INTERACTION_UE* is the interaction of each style variable with *UE*.

Results are presented in Table 5. In the columns labeled [1], the significant negative coefficient on the interaction variable of *WORDS_AL* and *UE* indicates that the length of earnings press releases diminishes the relation between abnormal returns and unexpected earnings, in support of *H2*. In the columns labeled [2], the significant negative coefficient on the interaction variable of *NUMERICAL* and *UE* suggests that the numerical intensity of earnings press releases diminishes the relation between abnormal returns and unexpected earnings, in support of *H3*. In the columns labeled [3], results of the regression including *COMPLEXITY* and its interaction variable suggest that textual complexity is a suppressor variable, i.e. textual complexity in earnings press releases suppresses the relation between *UE* and abnormal returns. These results do not, however, support *H4* since neither *COMPLEXITY* nor its interaction variable are significant. Although there is some evidence in support of hypotheses *H2* and *H3* considered separately, when the style variables are considered collectively, the results in Table 5 column [4] show that only the length of press releases appears to negatively affect the relations between abnormal returns and unexpected earnings. Both the variable indicating earnings in excess of analysts' expectations *MBE* and the *TONE* variable remain significant in affecting abnormal returns.

To address whether investor sophistication plays a role in the market impact of tone on the relation between unexpected earnings and abnormal returns (*H5*), a proxy for investor sophistication and interaction variables calculated as the product of the investor sophistication proxy and *UE*, and the product of investor sophistication proxy and *TONE*, are included in regression equation *(1).* The proxy for investor sophistication is the percentage of a firm's shares owned by institutional investors. Results, untabulated, do not show a relation between abnormal returns and the investor sophistication proxy or its interactions. Inclusion of the variables do not affect the relation between market returns and either *TONE* or *UE*. In other words, both unexpected earnings and tone of the earnings press release have the same market impact regardless of variations in the percentage of institutional owners in firms' shareholder bases.

VI. Conclusion

In summary, results reported here show evidence that tone of earnings press releases in addition to financial results affects market reaction to the earnings announcements. In addition, evidence suggests that the impact of unexpected earnings on abnormal returns is negatively affected by the length and numerical intensity of press releases. Results are similar, controlling for the sophistication of firms' shareholder base.

Opportunities for further research include an examination of whether the analyses and results reported here, which are based on observations from a single industry, generalize to other industries.

Additional research can also address further nuances of the firm-investor communication process examined here. It can be argued that the verbal portions of earnings press releases represent cheap talk (defined as "costless, nonbinding, nonverifiable communication" (Farrell 1987)). Since a rational investor would not be expected to respond to cheap talk, evidence of tone affecting the relation between unexpected earnings and market reaction might be viewed as evidence of irrationality – purely affective bias caused by the tone of disclosure, or limited attention causing an investor to focus only on the most salient feature. If the verbal portions of earnings announcements is relatively unconstrained, why wouldn't all firms maximize positive tone? One answer is that subtle communication choices might affect management credibility, whether immediately or over time. Consider, for example, the frank comments of Warren Buffet, CEO of Berkshire Hathaway, Inc. in the company's inaugural earnings press release (*Business Wire*, 2006)[12]: "…our insurance business has benefited in a major way from the absence of catastrophe losses. This is due not to managerial brilliance but rather to good luck."

An explanation of the results presented here is provided by prospect theory (Tversky and Kahneman, 1981, 1986), which predicts that framing financial performance in positive terms will cause investors to think about the results in terms of increases relative to reference points and to value the performance accordingly. Earnings press releases employ an array of such reference points including prior period earnings for all and/or part of the firm and time period, prior period levels of some metric other than earnings, or current period performance relative to future expectations. Evidence reported here suggests that tone might contribute to establishing and/or emphasizing reference points.

[12] Prior to this release on November 3, 2006, Berkshire Hathaway had historically not issued earnings press releases, but only announcements of the date on which financial results would be reported. This earnings press release follows Berkshire Hathaway's March 2006 acquisition of Business Wire, a distributor of corporate news, multimedia and regulatory filings.

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Figure 1
Word lists used to develop the tone measures for earnings press releases.

Positive tone (POSITIVITY) and negative tone (NEGATIVITY) are defined as the frequency count of the words on the following lists, scaled by the total word count of the press releases.

POSITIVITY word list

positive positives success successes successful succeed succeeds succeeding succeeded accomplish accomplishes accomplishing accomplished accomplishment accomplishments strong strength strengths certain certainty definite solid excellent good leading achieve achieves achieved achieving achievement achievements progress progressing deliver delivers delivered delivering leader leading pleased reward rewards rewarding rewarded opportunity opportunities enjoy enjoys enjoying enjoyed encouraged encouraging up increase increases increasing increased rise rises rising rose risen improve improves improving improved improvement improvements strengthen strengthens strengthening strengthened stronger strongest better best more most above record high higher highest greater greatest larger largest grow grows growing grew grown growth expand expands expanding expanded expansion exceed exceeds exceeded exceeding beat beats beating

NEGATIVITY word list

negative negatives fail fails failing failure weak weakness weaknesses difficult difficulty hurdle hurdles obstacle obstacles slump slumps slumping slumped uncertain uncertainty unsettled unfavorable downturn depressed disappoint disappoints disappointing disappointed disappointment risk risks risky threat threats penalty penalties down decrease decreases decreasing decreased decline declines declining declined fall falls falling fell fallen drop drops dropping dropped deteriorate deteriorates deteriorating deteriorated worsen worsens worsening weaken weakens weakening weakened worse worst low lower lowest less least smaller smallest shrink shrinks shrinking shrunk below under challenge challenges challenging challenged

Table 1
Descriptive statistics

Variable	*N*	Mean	Median	Std Dev	25th Percentile	75th Percentile
CAR [a]	1366	-0.010	-0.008	0.114	-0.073	0.054
UE [a]	1366	-0.006	0.000	0.235	-0.019	0.016
TONE	1366	0.568	0.600	0.247	0.415	0.750
NEGATIVE_TONE [a]	1366	0.007	0.006	0.004	0.004	0.009
POSITIVE_TONE [a]	1366	0.027	0.026	0.010	0.020	0.032
WORDS_AL [a]	1366	1969	1730	977	1307	2479
NUMERICAL [a]	1366	0.138	0.132	0.040	0.110	0.164
COMPLEX [a]	1366	5.242	5.254	0.247	5.090	5.405
lnMV	1366	6.223	5.935	1.997	4.804	7.455

[a] Data windsorized to the 99% and 1% levels to reduce potential impact of outliers.

Variables are defined as follows: *CAR:* Cumulative abnormal returns from day t-1 to day t+1, where day zero is defined as earnings announcement date. *UE:*EPS minus EPS reported in the previous year, scaled by beginning of period share price. *TONE = (POSITIVE_TONE - NEGATIVE_TONE)* divided by *(POSITIVE_TONE + NEGATIVE_TONE*); where *POSITIVE_TONE* is the count of positive words in text portion of press release scaled by total words in the text portion, and *NEGATIVE_TONE* is the count of negative words in text portion of press release scaled by total words in the text portion. *SCALED_TONE* is *POSITIVE_TONE – NEGATIVE_TONE. WORDS_AL* Total word count of the press release. *NUMERICAL:* Numerical terms in text portion of press releases scaled by total words in the text portion. *COMPLEX:* Textual complexity defined as characters per word. *lnMV:* Natural log of market value.

Table 2
Correlation between main variables
(Pearson correlations above the diagonal, Spearman correlations below the diagonal, with two-sided *P-values* in parentheses)

	CAR	*UE*	*TONE*	*NEGATIVE_ TONE*	*POSITIVE_ TONE*	*WDA*	*NUMERICAL*	*COMPLEX*	*lnMV*	*NIgt0*	*MBE*
CAR		0.036	0.098	-0.057	0.078	-0.016	-0.033	0.053	0.011	0.076	0.147
		(0.186)	*(0.000)*	*(0.036)*	*(0.004)*	*(0.561)*	*(0.224)*	*(0.051)*	*(0.694)*	*(0.005)*	*(0.000)*
UE	0.103		0.072	-0.037	0.056	-0.014	0.051	-0.011	0.054	0.078	0.065
	(0.000)		*(0.008)*	*(0.177)*	*(0.037)*	*(0.598)*	*(0.062)*	*(0.698)*	*(0.046)*	*(0.004)*	*(0.000)*
TONE	0.072	0.116		-0.834	0.606	-0.011	0.038	-0.005	0.212	0.166	0.140
	(0.007)	*(0.000)*		*(0.000)*	*(0.000)*	*(0.688)*	*(0.163)*	*(0.867)*	*(0.000)*	*(0.000)*	*(0.000)*
NEGATIVE TONE	-0.035	-0.058	-0.867		-0.187	0.036	-0.031	0.044	-0.071	-0.039	-0.086
	(0.190)	*(0.032)*	*(0.000)*		*(0.000)*	*(0.182)*	*(0.249)*	*(0.105)*	*(0.009)*	*(0.150)*	*(0.000)*
POSITIVE_TONE	0.072	0.127	0.613	-0.192		-0.024	0.098	-0.003	0.331	0.276	0.120
	(0.008)	*(0.000)*	*(0.000)*	*(0.000)*		*(0.369)*	*(0.000)*	*(0.913)*	*(0.000)*	*(0.000)*	*(0.000)*
WORDS_AL	0.002	0.026	-0.035	0.048	-0.011		0.028	-0.015	0.298	-0.071	-0.009
	(0.931)	*(0.337)*	*(0.190)*	*(0.075)*	*(0.677)*		*(0.308)*	*(0.586)*	*(0.000)*	*(0.009)*	*(0.000)*
NUMERICAL	-0.025	0.053	0.037	-0.018	0.077	-0.028		-0.659	0.175	0.133	0.017
	(0.347)	*(0.050)*	*(0.166)*	*(0.513)*	*(0.004)*	*(0.300)*		*(0.000)*	*(0.000)*	*(0.000)*	*(0.000)*
COMPLEX	0.044	-0.018	0.0003	0.015	0.008	0.005	-0.639		-0.033	-0.046	0.042
	(0.106)	*(0.513)*	*(0.991)*	*(0.5672)*	*(0.758)*	*(0.842)*	*(0.000)*		*(0.229)*	*(0.090)*	*(0.000)*
lnMV	0.028	0.040	0.228	-0.093	0.316	0.252	0.165	-0.039		0.304	0.139
	(0.307)	*(0.136)*	*(0.000)*	*(0.001)*	*(0.000)*	*(0.000)*	*(0.000)*	*(0.145)*		*(0.000)*	*(0.000)*
NIgt0	0.065	0.128	0.176	-0.053	0.280	-0.093	0.120	-0.044	0.311		0.138
	(0.017)	*(0.000)*	*(0.000)*	*(0.051)*	*(0.000)*	*(0.001)*	*(0.000)*	*(0.105)*	*(0.000)*		*(0.000)*
MBE	0.134	0.127	0.136	-0.089	0.134	0.008	0.023	0.036	0.143	0.138	
	(0.000)	*(0.000)*	*(0.000)*	*(0.001)*	*(0.000)*	*(0.781)*	*(0.3858)*	*(0.188)*	*(0.000)*	*(0.000)*	

Variables are defined as follows: *CAR:* Cumulative abnormal returns from day t-1 to day t+1, where day zero is defined as earnings announcement date. *UE:*EPS minus EPS reported in the previous year, scaled by beginning of period share price. *TONE = (POSITIVE_TONE - NEGATIVE_TONE)* divided by *(POSITIVE_TONE + NEGATIVE_TONE*); where *POSITIVE_TONE* is the count of positive words in text portion of press release scaled by total words in the text portion, and *NEGATIVE_TONE* is the count of negative words in text portion of press release scaled by total words in the text portion. *WORDS_AL* Total word count of the press release. *NUMERICAL:* Numerical terms in text portion of press releases scaled by total words in the text portion. *COMPLEX:* Textual complexity defined as characters per word. *lnMV:* Natural log of market value.

Table 3
Panel regression of event-window abnormal returns on unexpected earnings and tone
All regressions include fixed company and year effects.

Dependent variable: CAR

	[1]		[2]		[3]	
	Coef.	*P-value*	Coef.	*P-value*	Coef.	*P-value*
Constant	0.027	*0.50*	0.031	*0.46*	0.017	*0.67*
UE	0.033	*0.05*	0.033	*0.05*	0.032	*0.06*
lnMV	-0.012	*0.08*	-0.012	*0.08*	-0.012	*0.06*
MBE	0.026	*0.00*	0.026	*0.00*	0.025	*0.00*
NIgt0	-0.008	*0.43*	-0.008	*0.43*	-0.009	*0.41*
TONE	0.047	*0.02*	0.046	*0.02*		
WORDS_AL			-0.213*D-05*	*0.72*		
SCALED TONE					3.592	*0.00*
Scaled Tone^2					-59.178	*0.00*
N	1366		1366		1366	
F value &	1.29		1.29		1.30	
(p-value)	*0.001*		*0.001*		*0.000*	
Adjusted R^2	0.108		0.107		0.113	

[a] Variables are defined as follows: *CAR:* Cumulative abnormal returns from day t-1 to day t+1, where day zero is defined as earnings announcement date. *UE:*EPS minus EPS reported in the previous year, scaled by beginning of period share price. *lnMV:* Natural log of market value. *NIgt0* :indicator equal to one if net income is greater than zero. *MBE* : indicator equal to one if actual EPS exceeds average of analysts' estimates. *TONE = (POSITIVE_TONE - NEGATIVE_TONE)* divided by *(POSITIVE_TONE + NEGATIVE_TONE*); where *POSITIVE_TONE* is the count of positive words in text portion of press release scaled by total words in the text portion, and *NEGATIVE_TONE* is the count of negative words in text portion of press release scaled by total words in the text portion. *WORDS_AL* Total word count of the press release. *SCALED_TONE* is *POSITIVE_TONE – NEGATIVE_TONE. Scaled Tone*^{2} is *SCALED TONE* squared.

Table 4
Panel regression of event-window abnormal returns on unexpected earnings, tone and other stylistic variables.
All regressions include fixed company and year effects.

Dependent variable: CAR

	Coef.	*p-value*
Intercept	-0.098	*0.53*
UE	0.034	*0.05*
lnMV	-0.011	*0.09*
MBE	0.026	*0.00*
NIgt0	-0.008	*0.48*
TONE	0.048	*0.02*
WORDS_AL	-0.427*D-05*	*0.48*
NUMERICAL	-0.083	*0.63*
COMPLEX	0.027	*0.30*
N	1366	
F value &	1.29	
(p-value)	*(0.001)*	
Adjusted R^2	0.11	

[a] Variables are defined as follows: Variables are defined as follows: *CAR:* Cumulative abnormal returns from day t-1 to day t+1, where day zero is defined as earnings announcement date. *UE:*EPS minus EPS reported in the previous year, scaled by beginning of period share price. *lnMV:* Natural log of market value. *NIgt0* :indicator equal to one if net income is greater than zero. *MBE* : indicator equal to one if actual EPS exceeds average of analysts' estimates. *TONE = (POSITIVE_TONE - NEGATIVE_TONE)* divided by *(POSITIVE_TONE + NEGATIVE_TONE*); where *POSITIVE_TONE* is the count of positive words in text portion of press release scaled by total words in the text portion, and *NEGATIVE_TONE* is the count of negative words in text portion of press release scaled by total words in the text portion. *WORDS_AL* Total word count of the press release. *NUMERICAL* Numerical terms in text portion of press releases scaled by total words in the text portion. *COMPLEX:* Textual complexity defined as characters per word. *lnMV:* Natural log of market value.

Table 5
Panel regression of abnormal returns on quantity, numerical intensity, textual complexity, and interactions
All regressions include fixed company and year effects.

Dependent variable: CAR	[1]		[2]		[3]		[4]	
	Coef.	*p-value*	Coef.	*p-value*	Coef.	*p-value*	Coef.	*p-value*
Intercept	0.026	*0.53*	0.048	*0.26*	-0.137	*0.24*	-0.099	*0.53*
UE	0.136	*0.00*	0.152	*0.15*	-0.272	*0.54*	0.170	*0.79*
lnMV	-0.010	*0.12*	-0.012	*0.68*	-0.012	*0.08*	-0.010	*0.12*
MBE	0.025	*0.00*	0.026	*0.00*	0.025	*0.00*	0.247	*0.00*
NIgt0	-0.001	*0.36*	-0.001	*0.54*	-0.008	*0.48*	-0.008	*0.44*
TONE	0.041	*0.04*	0.051	*0.01*	0.048	*0.015*	0.046	*0.02*
WORDS_AL	-0.263*D-05*	*0.66*					-0.474*D-05*	*0.43*
WORDS_AL -UE interaction	-0.429*D-04*	*0.01*					-0.418*D-04*	*0.02*
NUMERICAL			-0.157	*0.24*			-0.086	*0.61*
NUMERICAL-UE interaction			-0.881	*0.05*			-0.714	*0.25*
COMPLEX					0.030	*0.14*	0.027	*0.30*
COMPLEX -UE interaction					0.059	*0.50*	0.011	*0.92*
N	1366		1366		1366		1366	
F value &	1.30		1.30		1.29		1.31	
(p-value)	*0.000*		*.000*		*.001*		*.000*	
Adjusted R^2	0.113		0.111		0.109		0.115	

[a] Variables are defined as follows: Variables are defined as follows: Variables are defined as follows: *CAR:* Cumulative abnormal returns from day t-1 to day t+1, where day zero is defined as earnings announcement date. *UE:*EPS minus EPS reported in the previous year, scaled by beginning of period share price. *lnMV:* Natural log of market value. *NIgt0* :indicator equal to one if net income is greater than zero. *MBE* : indicator equal to one if actual EPS exceeds average of analysts' estimates. *TONE = (POSITIVE_TONE - NEGATIVE_TONE)* divided by *(POSITIVE_TONE + NEGATIVE_TONE*); where *POSITIVE_TONE* is the count of positive words in text portion of press release scaled by total words in the text portion, and *NEGATIVE_TONE* is the count of negative words in text portion of press release scaled by total words in the text portion. *WORDS_AL* Total word count of the press release. *NUMERICAL:* Numerical terms in text portion of press releases scaled by total words in the text portion. *COMPLEX:* Textual complexity defined as characters per word. *lnMV:* Natural log of market value. *WORDS_AL -UE interaction: WORDS_AL* times *UE. NUMERICAL-UE interaction: NUMERICAL* times *UE. COMPLEX-UE interaction*: *COMPLEX* times *UE.*

Appendix – Affect disambiguation

Tone is a function of both affect and meaning, both of which can be altered by the context in which a word appears. This appendix describes the corpus-linguistics approach to disambiguation of the affect of the directional words included in the positive and negative word lists, also referred to as the tone thesauruses.

A generic issue in computational linguistics is the potential for polysemy (i.e., lexical ambiguity). A rich literature exists in the field of NLP dealing with word sense disambiguation. (See, for example, Preiss and Stevenson, 2004.) "Disambiguation" in NLP generally refers to sense disambiguation (e.g., distinguishing between use of the word "rock" to mean a type of music versus a piece of stone). Because earnings press releases are essentially a "sublanguage", i.e., with a limited domain of discourse, the sense of most words is fairly clear, lessening the need for sense disambiguation (Kittredge, 1982; Sager, 1986).[13] For example, in earnings press releases, it can be expected that the word "net" is more likely being used as an antonym to "gross" (although its part of speech may be an adjective, verb or noun) than as an item used by fishermen or hockey players.

In measuring tone of press releases, therefore, this study's focus is on *affect* disambiguation rather than *sense* disambiguation, specifically on distinguishing the positive versus negative context-dependent affect of the specific words included in the tone thesauruses. Many of the words in the tone thesaurus are relatively unambiguously positive or negative, for example "delighted" versus "disappointed". However, the directional words such as "increased" and "decreased" carry a particular potential for ambiguity and are the focus of this discussion. As an example, the word "increased" is included in the positive tone thesaurus, but if an earnings press release states that some generally undesirable item increase, e.g., expenses increased, then the word "increased" could not be considered unambiguously positive in this context.

To determine whether a particular directional word is generally used in a positive or negative sense, summary information for each of the directional terms in the tone thesauruses is obtained based on

[13] Kittredge (1982) examines a related sublanguage of newspaper reports of daily stock market activity.

the context of their occurrences in the corpus of earnings press releases. This summary information is examined to establish the percentage of occurrences in the context of inherently desirable or undesirable financial items described above. This approach is analogous to statistical disambiguation methods, reliant on a basic assumption: "that word senses are strongly correlated with certain contextual features like other words in the same phrasal unit" (Manning and Shutze, 1999, p. 250). Roughly, the approach is to disambiguate first on the strongest collocational feature and then to assign all instances of the ambiguous word to the majority sense in a document. This approach to disambiguation is not based on the assumption of independence of occurrences of words as with certain other methods (e.g., the naïve Bayes classifier explicated in Lewis 1998), but avoids the need to model specific interdependencies.

Context of words is typically described in terms of collocates. Words occurring near the target word are collocates.[14] The position of a collocate is designated by direction, i.e., to the left (L) or right (R) of a target word, and by words away from the target word. As examples, L1 refers the word occurring immediately to the left of the target word, and R2 refers to the word occurring two words to the right of the target word.

Empirical evidence shows that people can identify the sense of a word when given a relatively narrow collocate horizon, e.g., a window of ±2 words of context (Miller and Leacock 2000). The approach to affect disambiguation in this study uses a slightly wider collocate horizon of ±3 words in order to capture more context.

The 1,366 annual earnings press releases from Factiva (formerly Dow Jones) or Lexis-Nexus Academic for communications and computer companies during the years ended 1998 through 2002 forms the corpus for statistical disambiguation. The total number of tokens (individual occurrences of terms[15]) in the corpus is 3.3 million, and the total number of different words (also called types) is 26,109, after

[14] In this analysis, I count as collocates only those words appearing near thesaurus terms at least five times. A distinction should be made between the terms collocate and collocation. A collocation is "an expression consisting of two or more words that correspond to some conventional way of saying things" (Manning and Shutze, 1999, p. 151). If two or more words form a collocation, the meaning of the group of words is different from that of any individual word (e.g., "best practice"). In contrast, any word can be a collocate of any other so long as they appear near one another in a text.

[15] Terms include strings of numbers or letters.

excluding the 15 stop words[16]. As is generally the case in natural language processing, the distribution of occurrences of words is highly skewed (i.e., most words are rare). Within the entire corpus, only 2,158 words (8%) make up 90% of all the occurrences, and the remaining 23,951 words (92%) make up only 10% of occurrences. Only 58 different words appear more than 5,000 times, making up over 30% of all the occurrences; and 19,307 words appear 10 or fewer times, making up about 2.7% of all occurrences.

As noted, affect disambiguation in this study focuses on the directional words used in the tone thesaurus. Of the directional words included in the positive and negative tone thesauruses, the five from each that occur most frequently are shown in Table A-1. Panel A shows the total frequency counts of each word. Overall, earnings press releases include far more occurrences of upward directional words than downward. The relatively greater frequency of upward versus downward words in this corpus corroborates evidence from Rutherford's (2005) study of a corpus of annual report narratives that shows language biased toward the positive, a phenomenon referred to as the "Polyanna effect."

For each of the directional words in the tone thesaurus, the L3 to R3 collocates are sorted by frequency of occurrence and those collocates that make up at least 80% of the total are examined. Each collocate is categorized as generally desirable, generally undesirable, or neutral, based on their relationship with firm value in common equity valuation models In commonly used equity valuation models (e.g., the dividend discount model, free cash flow models and residual income models),[17] firm value is expressed as an increasing function of dividends, cash flows or income in excess of capital charges. Dividends and cash flows are generally an increasing function of income; and income is generally an increasing function of margins and revenues. Generally undesirable items, e.g., losses, expenses and costs, are those that reduce income.

[16] A common procedure in analyzing text, particularly in information retrieval, is to eliminate stop words (also sometimes known as "filter words"), which are words appearing so often as to provide little discriminatory power in information search and retrieval. For this analysis, I use a brief stop word list, including the following words: A, AN, AND, AS, AT, BY, FOR, IN, OF, ON, OR, THAT, THE, THIS, TO, $". Additional words appear on generic stop lists; however, because an accepted precedent does not exist for analysis of earnings press releases, and because one objective of this study is a detailed examination of the verbal component of press releases, the stop list used is intentionally brief.

[17] These equity valuation models are explicated in Stowe, Robinson, Pinto and McLeavey (2002).

For the upward directional words, the percentage of non-neutral collocates that are consistent with a positive affect are presented in Panel B. For the downward directional words, the percentage of non-neutral collocates that are consistent with a negative affect are reported in Panel C.

As an illustration, of the 9,682 L1 to L3 collocates examined for the word "increased," 3,118 collocates are non-neutral (i.e., they refer to generally desirable or undesirable items), while the remaining 6,564 collocates are neutral (i.e., neither an inherently generally desirable or undesirable item). Of the non-neutral collocates, 2,602 (83%) relate to a desirable item, most often "revenues" (695 times) or "revenue" (479 times). In comparison, 516 (17%) occurrences of collocates referring to a generally undesirable item were found, most often "expenses" (266 times). Similarly, of the 5,457 R1 to R3 collocates examined for the word "increased," 909 are non-neutral and 606 (66%) of these relate to a desirable item, most often revenues (143 times) and sales (98 times). To summarize, of the L3 to L1 non-neutral collocates, 83% of the occurrences (and for the R1 to R3 collocates, 66%) indicate that some generally desirable item increased, indicating that the word "increased" is appropriately included in the positive tone thesaurus.

In summary, this examination of the collocates of the most frequently occurring upward (downward) directional words indicates that the non-neutral collocates are, in the majority, consistent with a positive (negative) affect. An examination of the collocates of the less frequently occurring directional words (un-tabulated) gives similar results.

Table A-1
Data from corpus of 1, 366 earnings press release
Most Frequently Used Directional Words in the Tone Thesauruses and Percentage of Non-Neutral Collocates Consistent with Positive and Negative Tone

Panel A. Overall frequency counts of directional words in the tone thesauruses

POSITIVE: UPWARD DIRECTIONAL WORDS ↑	Overall Frequency count	NEGATIVE: DOWNWARD DIRECTIONAL WORDS↓	Overall Frequency count
INCREASED	6,173	DECREASE	1,531
INCREASE	5,936	DECREASED	1,238
GROWTH	5,035	DOWN	1,089
MORE	3,290	LESS	1,042
UP	2,517	LOWER	1,034

Panel B. Percentage of non-neutral collocates which are consistent with a positive affect

	L3 to L1 Collocates			R1 to R3 Collocates		
POSITIVE: UPWARD DIRECTIONAL WORDS ↑	Total examined	Non-neutral	Non-neutral collocates consistent with positive affect	Total examined	Non-neutral	Non-neutral collocates consistent with positive affect
INCREASED	9682	3118	2602 (83%)	5457	909	606 (66%)
INCREASE	4965	612	472 (77%)	6703	1503	1121 (75%)
GROWTH	7848	2073	2029 (98%)	5414	836	777 (93%)
MORE	3634	544	517 (95%)	4845	392	339 (86%)
UP	3168	530	530 (100%)	2577	161	110 (68%)

Panel C. Percentage of non-neutral collocates which are consistent with a negative affect

	L3 to L1 Collocates			R1 to R3 Collocates		
NEGATIVE: DOWNWARD DIRECTIONAL WORDS↓	Total examined	Non-neutral	Non-neutral collocates consistent with negative affect	Total examined	Non-neutral	Non-neutral collocates consistent with negative affect
DECREASE	1120	82	73 (89%)	1761	378	309 (82%)
DECREASED	1831	798	547 (68%)	922	71	32 (45%)
DOWN	1338	166	145 (87%)	1021	183	173(94%)
LESS	1117	411	278 (67%)	1478	217	99 (46%)
LOWER	1121	136	102 (75%)	1395	600	375 (63%)

Context of words is typically described in terms of collocates. Words occurring near the target word are collocates. The position of a collocate is designated by direction, i.e., to the left (L) or right (R) of a target word, and by words away from the target word. As examples, L1 refers the word occurring immediately to the left of the target word, and R2 refers to the word occurring two words to the right of the target word.